DAY 14 — Introduction to Facebook Prophet & Time Series Modeling

Day 14 focused entirely on the **Facebook Prophet** model — the algorithm used to forecast Tesla stock prices in the main project. Prophet is designed by Meta and is one of the most powerful libraries for time-series forecasting.

🔹 1. What is Prophet?

Prophet is an open-source forecasting tool that:

- Automatically detects trends
- Separates seasonality
- Handles missing data
- Provides uncertainty intervals
- Allows holiday/event adjustment

It is ideal for:

- Market forecasting
- Sales predictions
- Website traffic analysis
- Energy consumption forecasting

🔸 2. Prophet Model Structure

Prophet decomposes the series into:

✓ Trend Component

Long-term upward or downward movement.

✓ Weekly/Monthly Seasonality

Repeating patterns every fixed interval.

✓ Holiday/Events Component

Special events that affect data.

✓ Error/Noise Term

Unpredictable fluctuations.

This structure makes Prophet intuitive and extremely accurate.

🔸 3. Preparing Data for Prophet

Prophet requires two columns:

- **ds** → Date
- **y** → Target value

We converted:

df['Date'] → ds

df['Close'] → y

This standardized format helps Prophet understand time-based ordering.

🔸 4. Fitting the Prophet Model

We used:

```
model = Prophet()
model.fit(df)
```

Prophet automatically:

- Detects trend changes
- Identifies weekly seasonality
- Handles gaps in dates
- Learns patterns efficiently

🔸 5. Forecasting Future Values

We created future dates using:

```
model.make_future_dataframe(periods=30)
```

Then predicted:

```
model.predict(future)
```

The output included:

- Predicted values
- Lower & upper uncertainty bounds
- Trend line
- Seasonality curves

Key Takeaways from Day 14

- Prophet is the best model for time-series beginners
- Requires minimal preprocessing
- Strong for volatile datasets like stocks
- Trends and seasonality are automatically learned